

Learning Activity Sheet for General Science

Quarter 3

Lesson

7

Learning Activity Sheet for Senior High School - Genral Science
Quarter 3: Lesson 7

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LEARNING ACTIVITY SHEET

Learning Area:	General Science	Quarter:	3
Lesson No.:	7	Date:	
Lesson Title/ Topic:	Organ Systems Working Together		
Name:		Grade & Section:	

I. Activity No. 1: Major Components of the Immune and Lymphatic Systems

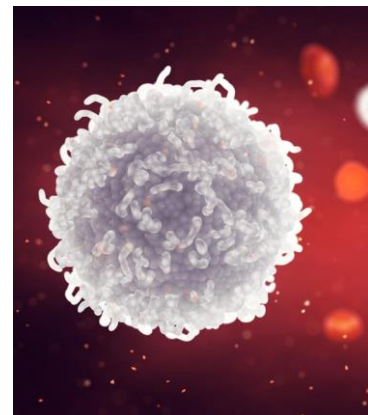
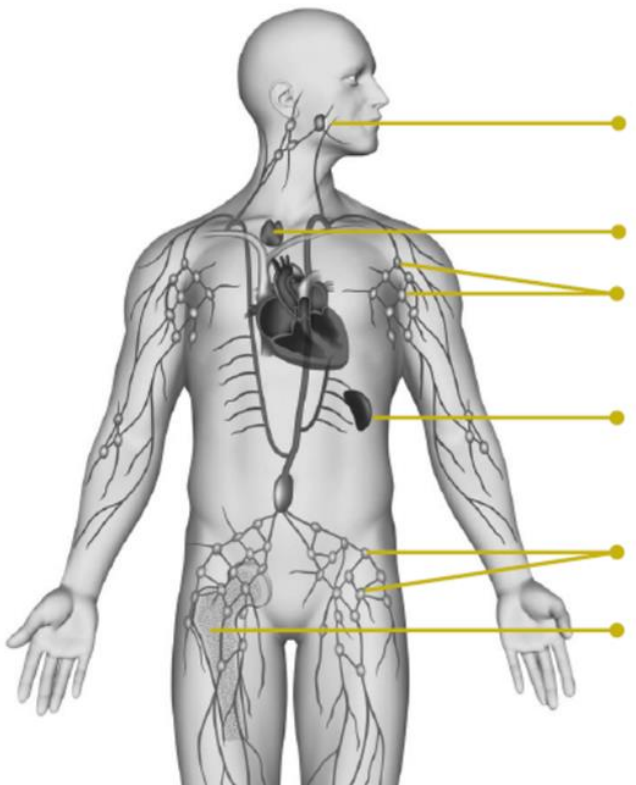
II. Objective(s): At the end of the activity, you should be able to identify and explain its major components and function of each system.

III. Materials Needed:

LAS

IV. Instructions:

1. Study the diagram of the immune and lymphatic systems .



2. Label the major components of the system. Use the **Word Bank** to **label** each part correctly

Bone Marrow
Spleen

Lymph Nodes
Thymus **Tonsils**

Lymphatic Vessels
White Blood Cells

3. For each component, write a short explanation of its function in the table below.

Components	Description/Function
Bone Marrow	
Lymph Nodes	
Lymphatic Vessels	
Spleen	
Thymus	
Tonsils	
White Blood Cells	

Note: You may refer to your references or visit the following site for your reference:

Cotterill, Simon. "Lymphatic System and Immune System | Medical Terminology for Cancer." *Cancerindex.org*, 2013, www.cancerindex.org/medterm/medtm9.htm.

Yox, Susan. "Understanding White Blood Cells and Differentials." *Www.patientpower.info*, 27 Feb. 2023, www.patientpower.info/navigating-cancer/understanding-white-blood-cells-and-differentials.

V. Synthesis:

Which component is most critical for producing the first line of defense against pathogens? Why?

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I. Activity No. 2: Immune System Relay – Defend the Body!

II. Objective(s): At the end of the activity, you should be able to:

1. Identify the major steps of the immune response.
2. Explain the major steps and the roles of specific immune system components of the immune response.
3. Explain how immune memory contribute to long-term protection.

III. Materials Needed:

- Station labels Task cards or printed station instructions
- Coins for circles (to draw the pathogens)
- Colored marker (for the anti-body tags)
- Stopwatch or timer

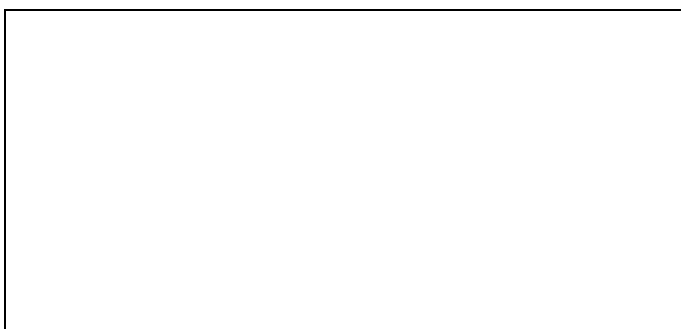
IV. Instructions:

1. Form 5 teams representing both invading pathogens and immune defenders.
2. There are five stations, each representing a stage of immune defense
3. Teams can start at any station to avoid crowding.
4. At each station, perform perform the following:
 - A. Draw or illustrate the assigned action
 - B. Explain the immune role, and
 - C. Answer the station questions before moving on.
5. The first team to complete all stations correctly wins!

Station Guide

Station 1: Invasion (Pathogen Entry)

Action: The “pathogen” enters the body through a cut, nose, or mouth.



Immune Role: _____

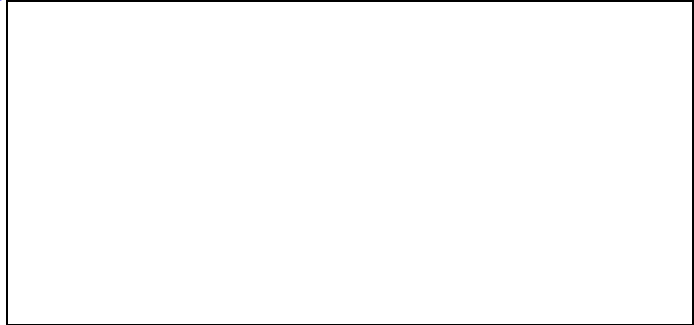
Station questions:

- Explain how macrophages identify and destroy pathogens.

- Why is the body's first line of defense important in preventing infection?

Station 2: Detection (Lymphatic Transport)

Action: The pathogen or antigen is carried through the lymphatic vessels to the lymph nodes.



Action: The pathogen or antigen is carried through the lymphatic vessels to the lymph nodes.

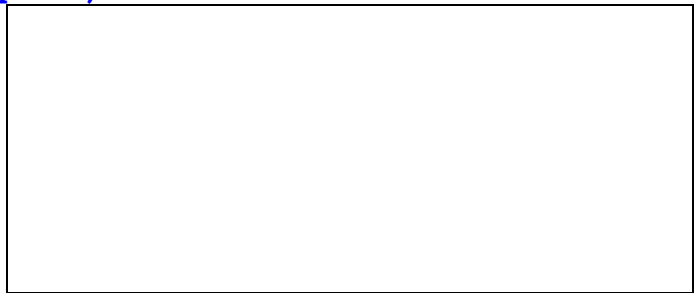
Immune Role: _____

- What is the function of lymphatic vessels in the immune response?

- What might happen if lymphatic transport is blocked?

Station 3: Activation (Lymph Node Response)

Action: Inside the lymph node, T-cells identify the pathogen and signal B-cells to activate.



Immune Role: _____

Station questions:

- Explain how T-cells communicate with B-cells to activate the immune response.

- Why is activation at the lymph node a critical step?

Station 4: Response (Antibody Production)

Action: B-cells produce and release antibodies that attach to pathogens. (Place an "antibody tag" to the pathogen using colored markers)



Action: B-cells produce and release antibodies that attach to pathogens.

Immune Role: _____

Station question:

- Explain how antibodies help stop infection and protect the body.

Station 5: Memory (Future Protection)

Action: B-cells produce and release antibodies that attach to pathogens.
(Using colored markers, indicate the memory cells)

Immune Role: _____

Station question:

- How do memory cells help the body respond faster next time?

- How do vaccines make use of this immune memory?

Suggested Rubrics for Grading

(The teacher may design his/her own rubrics together with the class)

	4	3	2	1
Worksheet Completion	The entire worksheet was completed.	3/4 of the worksheet was completed.	At least 1/2 of the worksheet was completed.	Less than 1/2 of the worksheet was completed.

Accuracy of information and explanation	Answer illustrates content accurate, a lot of thought, and preparation.	3/4 of the answer illustrates content accurate, a lot of thought, and preparation.	At least 1/2 of the answer illustrates content accurate, a lot of thought, and preparation.	Less than 1/2 of the answer illustrates content accurate, a lot of thought, and preparation.
Illustrations	Very neat. Illustration indicate accuracy of information	Neat. Most part of the Illustration indicate accuracy of information	Some part of the Illustration indicate accuracy of information	Few part of the Illustration indicate accuracy of information

Source: "IRubric: Worksheet Completion Rubric - SX34C4X," Rcampus.com, 2021,
<https://www.rcampus.com/rubricshowc.cfm?code=SX34C4X&sp=yes&>.

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I. Activity No. 3: Immune & Lymphatic Defenses in Action

II. Objective(s):

At the end of the activity, you should be able to:

1. Explain how the immune and lymphatic systems respond to specific threats.
2. Recognize the importance of immune memory and lymphatic functions in preventing illness.
3. Apply knowledge of immune responses to real-life scenarios.

III. Materials Needed:

Activity Sheet

IV. Instructions:

Match each case scenario in **Column A** with the correct immune or lymphatic response in **Column B**. Write the letter of your answer on the space provided.

Column A: Case Scenarios	Column B: Immune/Lymphatic Responses
_____ 1. Peter sneezed after dust entered his nose. The mucus in his nasal passages trapped the particles and prevented deeper infection. _____ 2. Maria accidentally cut her finger while chopping vegetables. The wound turned red and swollen as white blood cells arrived at the site. _____ 3. Jose received a flu vaccine. The antigens in the vaccine reached his lymph nodes where lymphocytes produced memory cells. _____ 4. Anna's tonsils became swollen while fighting a throat infection. The lymphatic tissue trapped pathogens and activated lymphocytes. _____ 5. Carlo had chickenpox as a child. When exposed again as an adult, his memory B and T cells quickly recognized the virus and prevented illness. _____ 6. A mosquito bite introduced parasites into Lea's bloodstream. Her spleen filtered the blood and activated immune cells to destroy the parasites.	A. First line of defense – mucus and cilia trap dust and microbes. B. Second line of defense – inflammatory response and phagocytosis at the wound site. C. Lymph nodes activate lymphocytes and produce memory cells after vaccination (third line of defense). D. Tonsils (lymphatic organs) trap pathogens and stimulate lymphocyte activation. E. Immunological memory enables rapid secondary immune response (third line of defense). F. Spleen filters blood, eliminating pathogens and activating immune cells.

Synthesis:

1. What happened in the given scenarios?

2. Which component of the immune or lymphatic system is responding to this situation?

3. How does the body's immune or lymphatic system defend against this threat?

4. Why is this response important for protecting the body now and in the future?

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I. Activity No. 4: Dress the Body for Defense – Vaccine in Action

II. Objective(s): At the end of the activity, you should be able to:

1. Identify major immune system components and their roles in defense.
2. Illustrate interactions among immune components after vaccination.

III. Materials Needed:

- Paper doll or body outline (one per student/group)
- Colored paper or sticky notes
- Whole bond paper
- Markers or crayons
- Labels: vaccine, lymph nodes, white blood cells, antibodies, memory cells

IV. Instructions:

A. Dressing the Body

1. Use the doll or body outline presented in the next page
2. Use colored paper or sticky notes to attach the following labels in the correct positions.
 - Vaccine
 - Lymph nodes
 - White blood cells
 - Antibodies
 - Memory cells



Source: Svagaa. (2018, June 11) Boy and girl are standing on a white background. Dreamstime.
<https://www.dreamstime.com/boy-girl-standing-white-background-boy-girl-standing-white-background-lot-kids-cartoon-image118467492>



Source: Svagaa. (2018, June 11) Boy and girl are standing on a white background. Dreamstime.
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B. Illustrate Interactions

Draw how the immune components interact after vaccination. Show the pathway of defense: white blood cells attacking pathogens, antibodies circulating, and memory cells preparing for future infections.



V. Synthesis:

Instructions: Answer the questions below based on your illustration.

1. What happens inside the body when a vaccine is introduced?

2. How do white blood cells, lymph nodes, and antibodies work together to fight infection?

3. Why are memory cells important in protecting the body from future infections?

4. How does vaccination prepare the body to respond faster and stronger against real pathogens?
